

2. By Newton's second law of motion,

$$F_{\text{thrust}} - F_{\text{resistive}} = ma$$

$$4.6 - F_{\text{resistive}} = 440 \times 10^{-3} \times 2.62$$

$$F_{\text{resistive}} \approx 3.45 \text{ N}$$

(ii) From the graph, it can be observed that as speed increases, acceleration decreases. This means that the resultant force decreases. Since the forward force (thrust) remains the same, it means that the resistive forces increase.

(c) From the graph, at $t = 0$, the acceleration of the car is about 8 m s^{-2} . In order for the car to propel vertically upwards, the resultant force must be greater than the weight of the car. However, the acceleration is lesser than g (9.81 m s^{-2}), which means that the force produced by the engine is unable to propel the car upwards.

2 (a) Thermal energy required, $Q = mc\Delta\theta$. Therefore,

$$Q = 5.6 \times 0.73 \times 1$$

$$Q \approx 4.09 \text{ J}$$

(b) (i) Since pressure is constant,

$$\frac{V}{T} = \frac{V'}{T'}$$

$$\frac{4.6 \times 10^3}{280} = \frac{V'}{281}$$

$$V' \approx 4.616 \times 10^3 \text{ cm}^3$$

$$\text{Therefore, } \Delta V = V' - V = 16.4 \text{ cm}^3$$

Note: For candidates who chose to calculate the number of moles n , it should be $\frac{5.6}{28}$, rather than $\frac{5.6}{14}$.

(ii) Work done on gas, $W = -p\Delta V$.

$$W = -1.0 \times 10^5 \times 16.4 \times 10^{-6}$$

$$W = -1.64 \text{ J}$$

Therefore, work done by gas is 1.64 J.

Note: Since the gas expands, work is done by the gas and hence should be a positive value.

(c) (i) The first law of thermodynamics states that the increase in internal energy of a system is the sum of heat supplied to the system and work done on the system.

(ii) Since the temperature change is the same whether the gas is heated at constant pressure or volume, the increase in internal energy is also the same.

Using the first law of thermodynamics, we have

$$\Delta U_{(a)} = \Delta U_{(b)}$$

$$Q_{(a)} + W_{(a)} = Q_{(b)} + W_{(b)}$$

$$4.088 + 0 = Q_{(b)} + (-1.64)$$

$$Q_{(b)} \approx 5.73 \text{ J}$$

3 (a) The principle of superposition states that when two waves of the same type meets at